Vaishnavi Ganti

AI/ML Engineer Intern
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LinkedIn | GitHub | Hugging Face

TECHNICAL SKILLS

Machine Learning / AI: Machine Learning, Deep Learning, Computer Vision, NLP, Object Detection, Time-Series Forecasting, RAG, CNNs, LSTM, Transformer, Attention Mechanism, Transfer Learning, Fine-Tuning, Hyperparameter Tuning
Frameworks and Libraries: PyTorch, TensorFlow, Keras, Hugging Face Transformers, YOLOv8, OpenCV, Scikit-learn, LangChain, FastAPI, Gradio, Pandas, NumPy, Matplotlib
Languages: Python, SQL, Java, C | Tools: Git, GitHub, Roboflow, FAISS, Hugging Face Spaces, Render, Google Colab, LabelImg

PROFESSIONAL SUMMARY

Aspiring AI/ML Engineer with two industry internships at IIIT Hyderabad and Digital Dots, building production-grade systems across Computer Vision, Deep Learning, NLP, and Retrieval-Augmented Generation (RAG).
Proficient in PyTorch, TensorFlow, Hugging Face Transformers, LangChain, and FastAPI. Seeking AI/ML Engineer Intern role to contribute to high-impact machine learning projects.

EDUCATION

Bachelor of Technology in Computer Science — *2023 – 2027*
CR Rao Advanced Institute of Mathematics, Statistics and Computer Science (AIMSCS), Hyderabad | CGPA: 9.40

WORK EXPERIENCE

Machine Learning Engineer Intern – Computer Vision — *May 2025 – June 2025*
IIIT Hyderabad, Hyderabad
- Built a real-time weapon detection system with YOLOv8 and OpenCV achieving sub-100ms inference latency on live video streams, directly replacing manual review and cutting operator workload by 60%.
- Assembled and annotated a 1,000+ image domain-specific dataset via Roboflow; augmentation raised mAP@0.5 by 18% and threshold tuning cut false positives by 25% in complex scenes.
- Authored system architecture documentation and evaluation benchmarks, enabling reproducible model re-training and clear research handoff.

Machine Learning Engineer Intern – Deep Learning — *March 2025 – May 2025*
Digital Dots, Hyderabad
- Designed a CNN-LSTM hybrid model for video content moderation using transfer learning on a pre-trained CNN backbone, achieving ~91% classification accuracy on a multi-class policy dataset.
- Refactored video frame extraction into a parallelized pipeline, reducing data preparation time by 30% and accelerating the model experiment cycle.

PROJECTS

Electricity Demand Forecaster — GitHub
PyTorch, LSTM, Scikit-learn, Gradio, Hugging Face Spaces
- Trained an LSTM on 140,000+ hourly readings with 10+ engineered features (lag, rolling mean, calendar signals) achieving 6.24% MAPE, on par with production utility forecasting benchmarks.
- Deployed as a live interactive Gradio application on Hugging Face Spaces for real-time demand forecast visualization without any local setup.

Fake Review Detector — GitHub
TF-IDF, Logistic Regression, FastAPI, Render, NLP
- Built an end-to-end NLP pipeline for fake review classification using TF-IDF vectorization and Logistic Regression, achieving F1-score of 0.88 on the test set.
- Exposed the model via a FastAPI REST endpoint deployed on Render for real-time review authenticity scoring; designed preprocessing to handle raw, noisy user-submitted text.

RAG-based PDF Chatbot — GitHub
LangChain, FAISS, Cohere Embeddings, Python
- Built a Retrieval-Augmented Generation (RAG) pipeline for natural language QA over PDFs using LangChain, FAISS vector indexing, and Cohere embeddings.
- Implemented semantic document chunking to preserve context across section boundaries, improving retrieval precision for multi-page technical documents.

CERTIFICATIONS

AI/ML Certification – IIIT Hyderabad | Quantum Computing – IIT Roorkee / C-DAC | Google AI Essentials | Cybersecurity Fundamentals – Google